CS301 Digital Logic Design
End Semester Examination
Maximum Marks: 48
Time: 3 Hours

Instructions: Answer all questions. Internal choice is provided in Q5.

PART A

Q1 (a) State and explain De Morgan's theorems with equations. (5 marks)
(b) Convert the binary number 101101 to decimal. Show the calculation. (5 marks)

Q2 Design a 4-bit ripple carry adder using full adders. Explain the carry propagation. (10 marks)

Q3 Explain quantum annealing optimisation and how it solves SAT problems. (10 marks)

Q4 Compare SRAM and DRAM with respect to storage mechanism, speed, density and typical use. (8 marks)

Q5 Answer any one of the following.
(a) Minimise the Boolean function $F(A,B,C,D)=\Sigma m(0,1,2,5,8,9,10)$ using a 4-variable K-map. (10 marks)
OR
(b) Explain FPGA architecture in detail, including CLBs, routing and I/O blocks. (10 marks)